

Shagun Shaw

Backend Developer

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SUMMARY

Backend developer with hands-on experience building and deploying production-grade Node.js/Express systems, including real-time architectures (Socket.io, JWT-based auth), and end-to-end cloud infrastructure on AWS (EC2, Nginx reverse proxy, PM2, CI/CD via GitHub Actions). Comfortable owning a service from local development through deployment and monitoring.

TECHNICAL SKILLS

Languages	Python, Java, JavaScript, TypeScript, C++, SQL
Backend	Node.js, Express.js, RESTful APIs, Socket.io, JWT Auth, Nodemailer
Cloud & DevOps	AWS EC2, AWS S3, Nginx (reverse proxy, SSL/TLS), PM2, GitHub Actions (CI/CD), Docker
Databases	MongoDB, MySQL, PostgreSQL
Monitoring & Tools	Winston, Axiom, Git, GitHub, Postman, VS Code
Other	LLM-based App Development, RAG

PROJECTS

QuizBlitz — Real-Time Quiz Hosting Platform | *TypeScript, Node.js, Express.js, Socket.io, MongoDB, JWT, AWS EC2, Nginx, PM2* GitHub | Live Demo (on Render/Vercel)

- Engineered Google OAuth 2.0 from scratch (no auth library), implementing a stateful JWT system with refresh token rotation, per-device session management, and secure httpOnly cookie handling.
- Built a Socket.io-powered real-time quiz engine with isolated in-memory room state, synchronized question broadcasting, and a server-side timer with a 2-second buffer to guarantee submission consistency.
- Designed a custom scoring engine with order-based point distribution, streak bonus multipliers, and auto-submit handling for expired responses.
- Deployed on AWS EC2 behind an Nginx reverse proxy with HTTPS/SSL and a custom domain, managing the Node process with PM2 for automatic restarts (later migrated to Render/Vercel after the AWS free tier ended).
- Built a GitHub Actions CI/CD pipeline for automated SSH deployment to EC2 using GitHub Secrets and IAM-scoped credentials; centralized Winston, PM2, and Nginx logs into Axiom for production monitoring.

SecureDrop — Ephemeral File Sharing Platform | *NestJS, TypeScript, MongoDB, Cloudinary* GitHub

- Built a backend processing pipeline using Multer memory storage and Archiver to aggregate and compress multi-file uploads into a single ZIP bundle.
- Implemented a scheduled cron job to purge expired file bundles from Cloudinary and remove corresponding metadata from MongoDB nightly.
- Enforced security constraints via custom validation pipes blocking restricted file types, a 50MB payload cap, and tiered rate limiting.

ACHIEVEMENTS

- Honoured with Medha Samman (Taaza TV) and Pratibha Samman (Prabhat Khabar), 2023, for being school topper in the ISC 12th board examination.
- Selected among the Top 30 teams (internal college round) at Smart India Hackathon 2024 for an innovative tech solution.
- Core Team Member, GDG on Campus TIU (DSA domain) — DSA Lead conducting sessions and mentoring peers in data structures and algorithms.

EDUCATION

Techno India University, Kolkata <i>B.Tech, Computer Science and Engineering</i>	Aug 2023 – Sept 2027 <i>GPA: 9.07</i>
Rajasthan Vidya Mandir <i>Higher Secondary (Science)</i>	Apr 2021 – Mar 2023 <i>92.7%</i>